AI-Supported Numeracy Teaching: session transcript

Blake Segler | Using AI to Enhance Literacy and Numeracy Skills, Module 3 | Monday 9 November 2026

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[0:00:04] Participant: Move.

[0:00:13] Participant: From Thailand. Yes. Uh-uh. OKOKI can see.

[0:07:49] Blake Segler: Good afternoon.

[0:07:52] Jane Sutherland: Like, like? Can you hear me?

[0:07:56] Blake Segler: I can hear you. Thanks, Jane.

[0:07:58] Jane Sutherland: Okay, I know that we've got some of our participants in the room, but I just thought I would quickly run through and just make sure that I had given you controls to be able to create your breakout rooms.

[0:08:09] Blake Segler: It looks like I can let me have a yeah.

[0:08:12] Jane Sutherland: It's a little room icon up across on the thing across the top.

[0:08:17] Blake Segler: Yes, no, this is looking perfect, I think.

[0:08:18] Jane Sutherland: Says rooms, and then you can, yeah, Blake's in managing rooms. It says perfect, excellent, excellent.

[0:08:23] Blake Segler: Yeah, yeah, it said I could take over that control from you and I went straight ahead and did so.

[0:08:26] Jane Sutherland: It took, because I put you in as someone to take over, that's right. Okay, fabulous. All right, have a great session, Blake. Thank you. See you.

[0:08:33] Blake Segler: Thanks, Jay. It's got some things that I probably liked. HUM. Just gonna put things in the chat.

[0:15:54] Participant: HUM.

[0:16:16] Blake Segler: HUM.

[0:16:18] Participant: Hello!

[0:16:49] Blake Segler: HUM.

[0:17:10] Participant: Mhm. ED.

[0:20:35] Blake Segler: The. It's. Okay, good afternoon everyone. I think we're still expecting a few more people, but I'm a big believer in also respecting everyone's time and really appreciate all of you who are here on time and many of you who got here a bit early as well. So I'm going to kick things off for today and a few more people obviously may join as I talk through. A big thank you also to those of you who have been here for a little bit and never start.

[0:22:10] Participant: ED. I don't know.

[0:22:16] Blake Segler: Oh, if we could just mute our microphone. Sorry, everyone. Just fine. I think that's... Yeah. Thank you. Thanks to those who have already had a bit of a go at these two little interactives, the Microsoft Farm and the Padlet that are also hopefully visible in the chat. If you haven't yet had a go at those,

[0:22:18] Participant: I love.

[0:22:37] Blake Segler: If you could just do those while I'm doing a bit of an introduction to this session, I'll use some of the results of those to discuss as we go through. And, like my colleagues, I am speaking to you today on the lovely Whadjuk Nyungabudja, which is to say the lands of the Whadjuk people of the Nyungar Nation are here in Australia, and would like to, like my colleagues, pay my respect to the elders past, present and emerging of those people, particularly as we go into this session. One thing that I'd like to reflect on in terms of Australia's history with its First Nations people is the extent to which following the colonial settlement in Australia, we had this great loss of knowledge and perspectives and culture of our First Nations people. And while we work towards reconciliation in our own context and trying to really foreground some of that knowledge, that's gone a bit more silenced as a result of the sort of dominant Western culture that sort of took over. Again, there's a lot that we are missing that adds to the rich cultural diversity and the strengths that come from Australia having the oldest continuous living culture on this planet. And as we start to look towards today's session and AI and all of the capabilities that it does have and the ways in which it stands to potentially enrich our teaching and learning, I'm reflecting on that and the importance that along the way when we use this tool, We don't inadvertently do so in a way that is likewise going to diminish or silence rich cultural perspectives and knowledges that might not be as aligned with some of those that dominate some of the training data. that these systems use and the importance as we go through this of being careful that we do safeguard the importance of those cultures and knowledges. Oh, sorry, if you just joined, if you don't mind just muting your microphone, thank you.

[0:24:52] Participant: Remove.

[0:25:07] Blake Segler: Lovely, thanks for that. And so this is particularly an area again where you will be the experts. I know some colleagues have said that previously of, you know, for your context in Indonesia and your culture, what it is really important to be preserving. throughout all use of AI. I'm obviously here to talk you through some of the capabilities and how we can harness it and how we can be mindful of that. But as I think my colleagues have said, it's up to you in the end to look at what goes in and what comes out and make sure that that's something that you are happy with as both an expert But also, again, as a, you know, in a sense, cultural guardian of the knowledges that you are, you know, promoting through your teachings. HUM. And so, for my background, if you're not aware, so my name is Blake. I think Craig mentioned that I'm a course Coordinator here at Curtin University, as well as being our subject area facilitator for mathematics.

[0:26:10] Participant: You.

[0:26:17] Blake Segler: And particularly, obviously, looking a lot at the use of AI in this space, which is an interesting place to be in. My background historically, like many of you, I used to work in schools, but in Australia, obviously. For myself, particularly in the secondary context, looking at both mathematics and sciences and particularly physics in that regard, while also working with our

primary schools, particularly through outreach contexts. And Back when I was still in schools and I was still a little bit more emerging, I probably would have been someone that was fairly opposed to using AI myself for a number of reasons that we will look at throughout today and you already would have been... looked into a bit in the online part of this session in terms of some of its limitations. But as you would have seen in that first concept in the online part of this module, since then, AI has really evolved in leaps and bounds. and so it's capable of a lot more than it used to be, and is certainly ubiquitous in the way that it's pervaded education. And certainly in the Australian context, we now have piloted programmes of AI lesson planners that are being used in schools. We know that our pre-service teachers who by training to come teachers are engaging with these tools, whether we would like them to or not. And I would say personally, I'm now very much in the camp that I would like our pre-service teachers to engage with these tools. But I want to again use them to do that in effective ways and critical ways where they're ensuring that their expertise is front and centre and that what they are producing with these tools is fit for purpose. And so that's largely what we're going to explore today, is ways to do that. I'll start to push through some of the slides now. Hopefully we've got most of everyone we're going to see today. So, to give you a bit of an overview of today's module. We're going to start very briefly looking back at this notion of numeracy and what it means in your context. And again, thank you for those of you that have already contributed to that discussion, particularly in the Padlet. Also looking a little bit at some of your comments in the Microsoft form and our readiness to engage in this and what we're hoping to get out of it. But the main three things that we're particularly going to look at are considering our role in evaluating what comes out of AI from a numeracy specific perspective. So delving into some prompts, thinking about how we can more carefully design them to ensure that what comes out of AI is fit for purpose, and also looking at how we evaluate those outputs to critically consider some of the elements of them that may or may not again be fit for purpose. We're then going to particularly get into the meat of today, which is starting to develop a fully resourced numeracy task for the context that we work in, or the context that your PPG students

[0:30:07] Participant: Mmh.

[0:30:15] Blake Segler: might work in. And so again, you would have started on this in the online section. So we're going to look at the two parts of that, of first deciding what a good task would be, and how we go about effective use of AI for ideation. and then how we use AI to really efficiently produce a suite of resources that's going to support us with that. I'm going to try and do a bit of a balance of a few things as we go through, and we'll see how well I do balance these. One is that there was obviously an online component to this that I'm hoping we've done before. And again, I want to really be respectful of everyone who has put in the time

[0:30:49] Participant: I HUM. No, no.

[0:31:04] Blake Segler: to make sure they completed that and try not to spend too much time rehashing that. If you haven't yet got around to that, there'll probably be some opportunities to try and catch yourself up or some opportunities to revisit it after this session. But I don't want to waste too much of people's time by just repeating what you would have explored.

[0:31:12] Participant: Mhm. HUM. No, no, I like them to play the game. Play, play.

[0:31:27] Blake Segler: Online. The other part, and if we can just cheque on microphones again, sorry.

[0:31:33] Participant: Thank you. Where should I go?

[0:31:36] Blake Segler: I think it's just over here. OK.

[0:31:38] Participant: HUM.

[0:31:40] Blake Segler: The other part is I want to get a bit of a balance, obviously, of where I am helping to learn my expertise in terms of providing some modelling and explanation of the different strategies we might use when engaging with AI. and also providing you suitable time to actually explore this in small groups on your own, and particularly to hopefully get at least a very decent start and close to the finish point of developing A numeracy task yourself so that you're not having to spend too much time

[0:32:12] Participant: Right.

[0:32:18] Blake Segler: after this session, continuing on to do that. But let me know how we go balance wise with that. In particular, again, as we go through, similar to my colleagues, if you want to ask any questions at any point, please do so. Just put something in the chat or do the little raise hand reaction thing and I'll call on you to.

[0:32:32] Participant: Okay.

[0:32:42] Blake Segler: ask what you'd like. And as we go into some of those more small group individual opportunities, again, let me know if you would like a little bit more clarity before we dive into that more independent practise, or if you need any assistance while you're working through those.

[0:32:55] Participant: No. Yes. HUM.

[0:33:01] Blake Segler: But otherwise, let's particularly jump into reviewing some of these elements. And we'll start with looking at the responses that have come through for that Microsoft form. So thank you to those of you that have had a chance to do that. I'm just going to put. that on my screen now. And hopefully you can see all of these results that have come through as well. Just give me a quick nod if you can see the Microsoft form up on your screen as well. Thanks, I can see Tamara nodding along. Okay, and so just see as a bit of a check-in. where everyone's at. Okay, so relatively confident in both our own and maybe to a slightly lesser extent, AI's ability to engage in personal numeracy. Okay, but perhaps with not incredible efficiency ourselves. Probably a bit likewise for our second one. This is likely to be a trend across all of these, I would say. HUM. Developing resources, again, fairly confident in both our own and AI's ability. with again starting to get a bit more of a reasonable chunk of people that are only somewhat comforting in their ability to do that quickly. Differentiation and scaffolding, which I would say in the Australian context, this is certainly one that a lot of our pre-service teachers, which be the kind of again your PPG students struggle quite a bit with, is being able to differentiate effectively and particularly manage their workload and being able to produce resources that allow for differentiation. And that's one where, nice to see, we are still feeling confident, many of us, but perhaps with a bit more of a spread down to those that aren't really confident in differentiating and scaffolding. But particularly this is where people are starting to really feel that it's time-consuming to produce. Assessing is an interesting one. Okay, so this looks like our first one in which no one's feeling completely confident in how they're assessing numeracy, but otherwise fairly similar. But we are, interestingly, I think more confident in how quickly we can do it, which is a bit of an interesting find. Um, and then, uh, looking, oh, it's just updating as more people contribute. You can see the last couple on our screen. Again, looking at some of these

pedagogical considerations of explaining ideas, choosing representations. Again, relatively confident, but for some people, not particularly time efficient, and likewise with feedback. we feel we can do it, but can we do that efficiently? And within again the Australian context, so again, I'm predominantly A secondary background and I think I saw from your pre-programmed surveys, the majority of you were as well. We have up to 32 students in Western Australia in our secondary classrooms. So being able to give effective feedback to all of them is very difficult to do efficiently. And so those will provide us a few good places to explore as this session goes on. Thank you again for your contributions to those. And as we start to consider this, I'm just going to refer back to again the grounding session of the online section. So in the online part of this module, you already explored again that numeracy means different things in different countries. is enacted differently in different countries, is different people's responsibility in different countries. And why this matters again is, as Craig mentioned back in his introductory session,

[0:37:42] Participant: No.

[0:37:42] Blake Segler: What AI draws on as its training data is going to influence what the AI outputs. And what we need to be particularly mindful of is, again, if we're using an AI system that is trained on a lot of data from a context very different to our own, then the outputs might not actually reflect our own context and culture. HUM. And so, again, if you had the chance in the online section of this to consider those three case studies, you looked a little bit at Australia, Japan and South Africa. Really important to compare and contrast that to the Indonesian context and think about what's going to come out from those.

[0:38:37] Participant: Anya.

[0:38:39] Blake Segler: Yes. So I'm not sure if that was someone trying to speak then. As I can see as well, we've got a bunch of posts in the tablet. Thanks for these. And these align quite a lot as well with what Yuli was talking about back in our very first introductionary brief session of how we're looking at numeracy in the Indonesian context and the importance, again, of distinguishing it from mathematics. Okay, that it's not simply calculation. It involves this deep reasoning and consideration of real world contexts across a range of different elements. And so you want to use that to guide as we go into our work. HUM. how we actually again construct our prompts to get effectively effective numeracy tasks that are going to be fit for Indonesian context and then evaluate the outputs. HUM. And so, just give... A minute or so to, if you haven't yet, have a bit of a read of some of the other. responses that people have posted to the Padlet. You can do it on the screen that I'm sharing or you can get up your own web browser. Also an opportunity, if anyone would like, if you had had a chance to do that online module and look at some of the different context of numeracy and you had any reflections you wanted to share back. please do take this opportunity to do so. But otherwise, very shortly we're going to get into the core concepts of today's session. by just giving a minute's break or so from me speaking, if there's anything you'd like to share or to also compare some of the contributions of your cohort around your consideration of numeracy in your context. HUM. We're just going to have about 30 seconds or so, which is not long, because you can review these afterwards as well, or in one of our breaks. Again, if there's anything that we do on the show as a reflection, please also do feel free to put something in the chat or to turn your microphone on to contribute. But otherwise, in another 30 seconds or so, we'll particularly get started. I. Okay. Now, a very last little bit of readiness before I get into a fair. Oh, sorry. I can see Sonja has his hand up. Apologies if I've mispronounced your name there. Please go ahead, turn on your microphone.

[0:43:15] Participant: HUM. Yeah, thank you, Mister Blake. Well, I want to share my experience in as a teacher in my school, professional high school. You know that most of my students, I can say that So... They don't like math; they are afraid of math, so when the teacher comes to the class, so they... they don't know how to do. Okay, now, you know that nowadays that there is an AI, so they can do everything, they can ask everything, they can solve everything with AI. But when I ask them what is the concept of numeracy, They also do not know, so the AI only they can ask everything, but they don't know the concept. Hey, so... Do you have some trick to do to what is it to give, motivate them, how to love the numeracy and so on? Thank you, sir.

[0:44:34] Blake Segler: No problem. Could I just clarify, are you talking about your PPG students or are you talking about students in your schools?

[0:44:49] Participant: Yeah, yeah.

[0:44:52] Blake Segler: Which one? Sorry, was it the PPG students or students instance?

[0:44:54] Participant: Yeah, the... A student in school.

[0:44:58] Blake Segler: Ah, wonderful. Yeah. HUM. So I would say a few things. First of all, I think what you're asking particularly gets to the heart of what we do mean by numeracy and its distinction from mathematics, of a large part of what we are aiming for when we talk about numeracy isn't a reliance on high level abstract mathematics. It's not a reliance on a procedural way of calculating through things. It is that personal capability to approach real life contacts. with confidence when those contacts do involve some level of mathematical literacy. So wherever that is, for example, managing my personal finances or when I'm out of the shops in that sort of financial context, Whether that is when I engage with news and media and I say things like statistics presented towards me. I want to be able to go into those scenarios and be able to confidently make sense of that information myself, and you know, similar to really the technological advance of the calculator, which is always the one we like to use as a bit of a... parallel. Yes, I can use a calculator to answer many problems, but I don't really want to be pulling one out in the middle of my local market in order to calculate costs of things. And I don't want to, every time I engage with a news story, pull out a laptop, open up ChatGPT and ask ChatGPT to try and make sense of things for me. Lots of the time I find myself even just having conversations with friends and colleagues, okay, where we'll be engaged in discussions with each other. And those discussions, again, involve a level of personal literacy that Tamara is going to talk about in Wednesday session. Or often they do involve a level of personal numeracy that again, I want to be able to actually myself comfortably engaged as a human in that conversation without having to pull up these tools. I'm the really big one that's been happening over the last few days, for example. Many of you may have been following the midterm elections. in the United States. And there is a wealth of numeracy elements to explore in that in terms of the different ways that, you know, the electoral map is shown and that critical numeracy to be able to compare, you know, maps that are prioritised based off of land area. compared to voting populations and to understand the nuances of where electoral boundaries are drawn and the difference between, you know, popular votes and all those sorts of things. There's a lot of numeracy involved in being able to understand real world important issues that are occurring around us. And while again, we may still, as part of that, use AI as a tool, in the same way that for some of these elements, I'm going to just in general use the internet. as a tool to search some of the nuances of American politics that I might not be aware of myself when I explore this. I'm still going to use that tool, but I want to have, first of all, a degree of personal numeracy that would allow me to just

inherently engage in the topic. and also sufficient personal numeracy that when I do use that tool, I'm able to assess whether or not what comes back seems right or not. And again, to use that calculator analogy, for many years now, we've been talking through that idea of the sanity cheque with calculators, or when I type in a calculation and I get a result out. Do I believe what's actually come out of my calculator? Because it's very easy to make mistakes using calculators, even more so if you're using software programmes as well as your next step beyond that. When I use AI in particular, and we'll talk about this in a moment with some of the example prompts you explored. I actually need some additional levels of critical evaluation to cheque that what comes back is actually what I am expecting. So that would be my first part of my response to you. The other two parts. is obviously actually exploring the ways in which numeracy is relevant. And we can do these in a couple of ways. I particularly like student-led discussions around relevance. Okay, so I find that as a mathematics teacher myself, one thing I would often do with topics, and I know we are now going to divert a little bit for a moment, but rather than simply telling students what the applications of a piece of mathematics was actually involving them in the discussion of seeing if they can identify real world contexts in which that would be relevant. Now, we don't have that as much necessarily in numeracy teaching, because we will front and centre be looking at relevant contexts. But what we might find still is that some of the numeracy contexts we may choose, and we'll talk a little bit about this. again a little bit later this session. But while they may be real world relevant contexts in general, they might not be as personally relevant to the students that we're working with, whether that's because of just the diversity of students, within our classes and the reality that we're working with so many people at one time, or whether it's because of their stage of life, in that sometimes we look at things like taxes and bank accounts and investments and things like that, which will be relevant in future to students. But perhaps not, you know, at the stage of life that they're currently at. but we can engage in some conversations around where we might still use similar processes that are more meaningful to them. But the other big one, and it will be, again, one of the big focuses of this session, so really glad for this question, is we actually just need to choose tasks that are going to convey that relevance. And sometimes we fall a little bit into a trap when we do this. And again, it is a large part of the second concept that we'll talk about today. But sometimes I find particularly even with AI, I will ask for examples of problems that are going to be relevant to my students. And I'll look at what's come back and I'll think, if I was a, let's say, 13 year old, and this is a problem that was put in front of me, I don't think I would find it relevant at all, and I don't think I would meaningfully engage with it. And being able to be critical about some ideas of numeracy tasks and push through to find what students are going to engage meaningfully with and see the purpose behind. That's really important and that is again somewhere where AI can actually be a great help. So we'll explore that a little bit more as the second concept that we look at today. But a really great and pertinent question. Thank you so much for that. So again, if there are any other questions or comments, please do feel free As we go through, just put up your hand or put something in the chat and we can address it. I did also... Oh, yes, Amalia, please go ahead. HUM.

[0:54:27] Participant: Yeah, thank you so much, Mr. Blake, for giving us a lot of knowledge about it. Well, this is what I have done to my PPG students when I asked them to modify the questions. Actually, I teach English and at that time I asked my PPG students to

[0:54:33] Blake Segler: HUM.

[0:54:51] Participant: modify the text before using it to teach it to the students and then there is a little bit difficult for them to modify since they use AI directly without giving the specific prompts and The text usually they make because at that time I also ask them to, what is it? Reading text here for Teka in Indonesia. Here there is test of academic competency and I asked them to modify. And involve the numbers, involve the numeracy in the text and usually mostly they tend to use a long text to make it more and more difficult and I asked him to modify just two or three paragraphs and then don't forget to. What is it to take the text from? such as the government website from Ministry of what is it? Ministry of Economics and etc. and they can take the table and sometimes they get confused about it and They just prompt in AI as simple. Just write, please, you make blah, blah, blah, 5 numbers. talking about blah blah blah and I think they have to revise it again and well, I'm not sure about the ability of my PPG students modifying the problems becoming literacy texts. So please, Mr. Blake, to give us the the example how to guide them easily to modify the text become literacy and numeracy and how to make a good prompt to my PPG students. Thank you, Mr. Blake.

[0:57:04] Blake Segler: HUM. Definitely. So that one I might just part for now and circle back to also in the concept too when we look at the prompting.

[0:57:09] Participant: Yeah.

[0:57:19] Blake Segler: But I will specifically try and construct one for a more literacy context of how we might go about producing a strong prompt to produce some good texts that also involve literacy. Um, so I... We'll remember to do that in concept too, but if I don't, then please, Amalia, just let me know again and we'll focus more on it when we get to that. Wonderful. Uh and so just uh also playing a little bit of pre-survey uh pre-programmed survey data as well. It was very interesting to see what some of our response were. It's great to see how many of us are engaging well with AI. Interestingly, both across those of you that do have numeracy backgrounds and those that don't, we seem to think... that AI would have less utility for teaching numeracy and less faith in its ability to do so. And one of the, for better or worse things, which I think is probably quite good, is that one of our key concerns was about this weakening of critical thinking. And that is a theme that we're going to keep coming back to throughout this module and I'm sure the other ones as well. is while AI has the ability, if we use it ineffectively, and I would say unethically as well, that it can sort of diminish our critical thinking. Really, if anything, what AI does require of us is in fact a greater level of critical thinking. Because we really need to give a very concentrated evaluation of the outputs of AI in order for them to be fit for purpose. And so one thing we do want to foreground a lot again is not that we are using AI to replace critical thinking. We're going to use AI to replace a lot of the general low-level workload associated with the tasks that we would do. The things that would take us a large amount of time, but wouldn't really involve a lot of critical thinking and largely administrative sort of tasks, but then we actually need to be doing a lot of critical thinking ourselves, and we'll return to that. Concept and shortly. So first concept that again you would have looked at a bit in the online module. is looking at how AI approaches numeracy-based questions as a little bit of a cautionary tale. And this builds again on some of what Craig did with you about the general ways that AI approaches different problems, but in numerous it's more interesting. And so you would have had in the online part of this module a few different prompts to try out. And so your first one was just actually putting in some numbers for a large calculation that interestingly, if you didn't have a chance to have a go at, you likely would have found it gives the wrong value. And often if you compare, as it suggested a bit later, Trying that another time of with a colleague that's also done this, it doesn't even give a

consistent value. More interestingly, we might jump straight to prompt two, which is one of my favourite prompts that I've devised for this session. because this is the prompt where I have purposefully given it. Numbers are written with the Indonesian conventions. And of course, in Indonesia, the convention Friday numbers is pretty much the complete opposite of the United States, where we would expect a lot of the training data for AI to come from. And that means that when I put this into an AI, I like to for modelling. use private browsers so that it doesn't draw upon my chat history and it's a little bit more neutral in the responses it gives out. When I use browsers that I've already been working in, then it's obviously remembering everything we've done before and so it starts to just... go along the same lines of thinking or particularly panda to, you know, styles of teaching that it knows I like. So this is just a little bit closer to probably what we'd see. What you would see if you do it yourself. Oh, that is all right. Okay, and so you can see again, as you would have done, okay, that in attempting to try and reconcile, you know, the periods and the commas, eventually it's just decided to change my 2300.456 into 2 million, 300,456. and given a completely wrong answer. And this is where again we might try to rectify that a little bit with counting. So I might specifically tell it, note that I am using the Indonesian. Manson. On numbers. Now, this is interesting. I'm going to see what comes up today, because I've done this a few times over the course of just this morning and gotten out some different results. So we'll see if it does get it right this time. The first time I tried this, I got it completely wrong. This time. And it certainly hasn't used the correct Indonesian conventions in giving... Oh, no, sorry, actually, that's fine. And has it done it correctly till 800? That one is correct. Lovely. When I did it this morning, it managed to get it even more wrong. than it had done the first time. So again, there's a bit of probability sometimes in the outputs that we've got to be careful of. But this is otherwise a nice example of where I might be able to overcome the limitation of AI if I give a bit more context. But I do still need to be careful about if I have... given enough context for it to actually get things right. Because when I do... just the basic prompt, okay, and it's trained in a context different to your own. It can, even for something really simple, like a straight sum of 2 numbers, it can actually misinterpret and get that wrong. The third prompt that we had is the one that I particularly want to talk about that's going to be very relevant to. a lot of what we do around numeracy. And so this is the prompt that I had for the online part of this. Our two shops, one offering rice at 3 kilogrammes for 4 million rupiah and one doing 5 kilogrammes for 9 million rupiah and looking at which is the better buy and how We might explain that. And if you've done the online bit, or again, even just listening to this, there's at least one thing that should be very concerning with that prompt that is purposely embed. which is how realistic these values are of 4 million rupiah is just an insane amount of money to be spending on only 3 kilogrammes of rice. HUM. But every single time I've tried this in any AI system that I've looked at, ChatGPT, Claude, Copilot, all of these different types of AI, not a single one has commented to me that that might be. B and history. I think just because I've got this Teams tech going alongside, it's slowing down my internet a little bit, but here it's come through. And so again, not a single consideration about whether this is realistic, which again, for someone who does have reasonable numeracy. I would expect anyone with reasonable numeracy to first of all say, oh, no, I'm not going to buy from either shop. That's way too much money. Or to at least look at these in question, are those actually correct or is there a mistake in the numbers that I've been given? But there's a bigger issue with this one in terms of numeracy. And I don't want to put people too much on the spot, but one of your prompts was to consider, again, whether this is the type of reasoning we would particularly want to see from a neuracy perspective. And again, rather than putting people on the

spot, I think I'm just going to say, as someone with a background in numeracy myself, obviously, that this is very much not what I would expect someone to do if they were engaging strongly with numeracy. My expected response from a highly numerate individual, particularly going back to sort of your point as well, Sanchezman, of, you know, this is meant to have personal capability associated with it. And again, when I put myself in the position of someone actually at the market going to buy rice, There's no way that I am trying to divide 4 million by 3 and 9 million by 5 and then compare unit costs. particularly when what I would expect is a very simple reasoned response to this, that if I bought two of the three kilogramme bags, Well, 2 lots of 4 million rip here is only 8 million rip here. So I can spend less money than a 5 kilogramme bag and still actually get more rice because I will get 6 kilogrammes rather than 5 kilogrammes. So that In terms of the personal capabilities of numeracy, that is what I actually want my students to respond, as well as, again, you know, noting that really I probably don't want to buy any bags of rice at all from this shop because all of them are so ridiculously overpriced. but it's not what my AI is giving me. Okay, because again, my AI is going to look towards the data it's trained on and a procedural-based approach is going to feature in a lot more of those examples because the whole point of a procedural-based approach is you can apply it to any scenario and it will work. So it's trying to almost remove that context specific information that would lead to more well-reasoned approach. And so someone that's again going to try and use this for numeracy teaching. This is something I need to be particularly mindful of is will AI's approach. to solving problems and explaining problems actually align with what I want to say. Because if I just present this sort of method to students because AI generated her, then I'm actually not going to be helping too much strong numeracy skills. And again, I just went to your point, I think this is where students are going to disengage for a number of reasons. It's... Too complicated. OK, this is not the type of mess that we're actually trying to engage with with this sort of problem. And it becomes unrealistic, okay, because we have removed all of that actual reality of how I'd engage with this in a real world context and just shipped it back to the abstract mathematics. So we need to be really cognizant. Of that. And this is one that we can have a little bit of a play around with to start off with. If you can think of some ways that I might change my prompt in this case, that might help give me an output closer to some of the things I might want to say. Please do feel free to type into the chat perhaps some things that I might add or an entirely different way of writing this from the start. and we can try some of them out together and see how they fare and if we can get a little bit closer to what we might want to say. I can say it looks like a few people are typing on their keywords at the moment. So just give a moment to see what comes. Sorry. While we're waiting, I'll try and model a few. I will say as a heads up, not all of these ones are necessarily going to get us to where we won. Okay, because one of the key takeaways from this is still going to be that cautionary. So, Malia's talked about the critical thing of human and how to make AI relevant with daily life, so I can put that into my prompt if I want. And. So I can sign it a role. You are a critical. Thinking. Human, and I might even say with a high level of personal. Numeracy. HUM. Three shops of a rice at blah blah blah blah blah. You need to discuss. as you would in daily life. That was the other part of Amelia's, Amelia's sorry, suggestion. Which is? the better buy. I can see if it does any better. I have a feeling for this one it might not. HUM. And no, it still does basically the same thing. HUM. I might even tell her. OK, for the same ones. HUM. Please avoid overly procedural. Approaches and instead use. What are you saying? I will say that does any. Better.

[1:16:15] Participant: And.

[1:16:18] Blake Segler: and still using that same method. I can, because I've played around with this one a little bit.

[1:16:18] Participant: Cooke.

[1:16:27] Blake Segler: If I tell it specifically. Do not use a unit price. The Fed. but instead provide other raising. directly telling it that I don't want a unit price method. will often change the way it goes about. And this one actually did land it, which I'm a little bit surprised by, if I'm being honest, because when I try this again this morning, I didn't manage to get this exact reasoning, but this is particularly what I'm after again, that I can get a 5 kilogramme package for 9 million RPR, but I can get two 3 kilogramme packages for only 8 million. So 6 kilogrammes of rice. and an extra million repair saved. And I do wonder a little bit. if even in private browsing, private browsing doesn't mean that nothing gets saved, if you're not aware, and I have a feeling through some of the different ways I've child this, it's still somewhat learnt that this is the sort of response that I do once. HUM. Frequent responses that I did manage to get it to do instead was to look at something like purchasing 15 kilogrammes of rice in both ways. Okay, so comparing 5, 3 kilogramme bags and how much that would cost to three, 5 kilogramme bags and how much that would cost. And that was better. but not really as good and context specific as what I would like. I'm still not getting. You know, the consideration of the rice being too expensive in this as well, and what I am having to do is be. really quite directed with it. I'm telling it, for example, not to use the unit price method when really that's a decision I would like it to make more so on its own and for a student. when I'm going to actually do a problem like this. I want them to consider the different options they have and what would be most appropriate, rather than having explicit directions. I might briefly point out, while we're just working on this prompt, that one thing I can also do. is ask, you know, a meta prompt is what we would call it of asking AI's advice on how I would fix this. So I can say, for example, I gave the prompt. We'll just say the which one too, which is about five. Your output was overly procedural. And used a unit cost method, whereas I would have preferred. A more reasoned. Alright. using numeracy such as two, 3 kilogramme bags. Would cost less and give more rice. A 5 kilogramme. Bag. Cooke. The output also didn't notice. That this is in. Unrealistically expensive. Cost for price. How could? I have. Provided. I better prompt. In order to get the output. I want to HUM. Little bit fast, not perfect, but this is another good skill that you can sometimes engage with more generally throughout your AI use is actually, again, we call this meta-prompting, engaging with the AI to help it instruction how you would. Interact with it effectively. HUM and so I... I get out. HUM. Something like this. Now the issue, as you may notice in this case, is it's a little bit too specific to the exact problem. Okay, so I've told it what I wanted. It's gone as far as to literally specify, look at two, 3 kilogramme bags. and compare it to a 5 kilogramme bag. So when I actually go to try and apply this in other contexts again. This one is just too specific. HUM, and sometimes also I... using somewhat general ideas I can start to apply towards prompts. This is a way that I can still start to improve what I'm putting in. In this case, I will say HUM. This one. I found incredibly difficult to overcome the training data that it's built upon. And so this is a place where I think as a practitioner, I just need to be more mindful of these limitations of AI. and whether this is the sort of problem that I want to be relying on the pedagogies that AI is using, or if I'm going to use AI for different purposes, and if particularly I'm going to engage in teaching something like this, making sure I'm either directing AI exactly towards how I wanted to discuss problems, Or more so deciding that that's not an area that I want to use AI for. That is an area where I myself am obviously going to take care of the actual teaching element of it and instead

use AI for things like coming up with the problems themselves and tasks. and supports, which is what we're going to explore a fair bit more over the course of the rest of the session. This one, again, you would have found a lot of things AI is really good at in a new SC context, and I did say in the online section as well that AI is getting better and better at a lot of things. We do need to be mindful, and this, you know, this is another way that you can also introduce with your PPG students and with their own students. Why that critical thinking element is so important? OK, that what I am saying from AI, sometimes it's straight up incorrect, like in the first couple of example prompts. Sometimes it just doesn't. look at things the way I would want them to be looked at. In this case, particularly from a pedagogical point of view, so more relevant to your PPG students rather than your students in schools. But when you get to some of those broader considerations of applying numeracy and literacy and general AI use, those same considerations of yes, it can give you a response that looks very persuasive, very strong. But it's not necessarily drawing on the types of reasoning that we may necessarily want. And when it's doing those to particularly produce arguments, that can be quite problematic for us. I'll just have a little bit of look at some things that's come through the chat. I mean, talking as well. HUM. This. So I'm just going to pull out some of the more pertinent ones. Amelia's asked, does this mean that every time I want to ask an AI a question, we have to give it as complete instructions as possible according to what we are looking for? And my answer, we'll look at this a little bit more as we go through, but I would largely say no. Um, and more and more, sorry, back. Wanna say six months ago, even there was a lot of conversation around effective prompt engineering, and that's still something you wanna consider. Craig talked about it a bit. We're still gonna talk about it today, but the reality is for most contacts. AI doesn't need as careful prompt engineering as it used to. And often I can put in really simple prompts and get out something good. But then I need to be quite careful when I go to evaluate what comes out. And based on what comes out, I'll often refine the prompt that I put in. And generally there's a balance. as to how much time I want to invest at the start, making a really good prompt to hopefully get something out that's more fit for purpose versus how much time I might need to then spend looking at what comes out and fixing it. And there's often a little bit of a back and forth. between the two of based on what comes out, I might change the prompts that I originally did, either within the same chat. Or in an entirely new chat, but we'll look at some examples of that in just in a moment. Let's answer your customer. Yeah, and Madeleine commented on that a bit as well. OK, that I'm looking for better ways of prompting. The puzzle is in the screen when you demonstrate. Ohh, I thought... I've got a comment from someone about seeing the screen when I'm demonstrating. I thought the screen should be visible. Can everyone see my ChatGPT screen at the moment as I'm doing this? I hope so. Yes, okay, good. HUM, thanks for that. So hopefully you can see that. I can also send through any more examples of these as well, if it's a little bit hard to see. Don't worry too much about reading every bit of text that comes out of these. You're in a moment again going to start exploring some of these yourselves as well, and you are to see a lot. Clear. And yeah, as my explainer said, starting to guide a little bit more how we might improve some of our prompts. The last one I do want to briefly talk about before we particularly get into some practical use of this and starting to actually build up some work product ourselves. is we're looking at the sorts of problems that are generated. Um, and we're going to do a favour on this, so we might not spend too much time. But you would have hopefully looked at this a little bit. yourselves. I might get up one example quickly for us to discuss a little bit together. HUM. We've got an interesting mix, by the looks of it, of both Bahasa Indonesian and English that it's decided to come out with. So this one that it has looked at as an example for a Kelly

three child. He is looking at everyday shopping in a traditional market in Indonesia. HUM. And it'll be up to you to look at the specific things that it's suggested that you can get and how much they cost and how much they are, and some of the questions that follow. So this, we've said it a few times that often you'll be the experts on this. I'm going to be very clear that I'm not entirely sure. what the vast majority of this means myself. Um, but... A little bit of a reflection and consideration of if we're looking at your Kelly three children. Do we think that this scenario that it's come up with is, oh, thanks, Magdalena, already. So Magdalena said that these are not appropriate prices either. What about the context? Do you think that a student in... That sort of age group. Are they gonna find this example meaningful to them? Or is that not really the sort of thing that they're likely to engage with? What do we think? I'll try and zoom in a little bit more.

[1:31:15] Participant: It's meaningful. The contact is meaningful, Blake. But the prices, as I said, unrealistic because children nowadays, I mean, Indonesian children, even though not all, I mean, they know the prices of them. The the fruit.

[1:31:35] Blake Segler: HUM. Which is good, HUM, and the sorts of conversations we want to have with them.

[1:31:38] Participant: Yeah, yeah, yeah, yeah. Pheasant is usually usually much cheaper.

[1:31:47] Blake Segler: HUM. And so some of the things, again, that I'd be looking at as well, again, are right. Well, do I think that it's going to be meaningful to all my students, or is it going to be meaningful to some, but a lot will not really be familiar with this? And this is a nice enough context that probably it is meaningful to a lot. I possibly, and this is where again, someone might have to translate a little bit for me, But I'd sometimes ask if the situation I'd got is a little bit too contrived of these three specific items with their amounts and prices. Is that what it really looks like when I go to a market and I'm trying to achieve whatever my everyday shopping task is? Because in a market, I'm going to have lots of different things I can buy and lots of different sizes and prices and things like that. Um, and so, while there might be... an element of the numeracy involved in visiting a market in this. This, I'm going to keep coming back to a little bit of that lovely question from Sonja Smith at the start of this. Sometimes when we... or either sort of problematize something and make it this nice little box around it problem of only these three things, only these amounts, only these prices, students will also call out that that's not really what things look like in there. Daily lives that they don't have this very specific example. they've got a lot more opportunity for, which is what we really want, that critical thinking and reasoning, because they're actually faced with a lot more openness and a lot more options. And again, what we're potentially doing with a problem like this, and again, I can't say fully because with apologies, I can't quite read the Vahas Indonesian. But sometimes when we get to specific, again, we strip away that. real world context and the actual numeracy skills, and we put it into a little bit that almost maths world of. No, here is just a problem to solve with a bunch of procedures and calculations. Oops, I just dropped something. And there's going to be a potentially specific answer. that we're looking for, which is not really how real-world numeracy works. And so that's where we're going to start to get towards a little bit more of what we're trying to aim for with our numeracy problems, what our workflow will look like, and then we're going to start to actually generate some much better tasks and work through them. No. And sitting behind all of this is that same concept that I know Craig's talked about quite a lot, and I'm going to read out quite a lot, is this notion of human in the driver's seat. Okay, and the critical role that we play as users of AI, and this is where that critical thinking comes in. of ensuring that we very much own what comes out. And

there's two elements in that. We'll break this down a little bit more in a moment. but it does speak a little bit towards, I'm just going to find, it was Amelia that sort of asked this about how much we need to prompt AI. But the two big parts we tend to consider are what goes in and what comes out. So what guidance have we given the AI about what it is we do want? And then what comes out? How are we carefully checking that to see that it's accurate and that it hasn't made mistakes, which it... Actually, quite readily does a lot of the time. that the pedagogy and particularly reasoning, you know, that it's using are appropriate for what we're wanting to say. and that it has actually gotten those important contextual elements. Whether they're simple things like that we're using realistic prices for what things cost in Indonesia, whether it is more of the cultural elements and the meaningful everyday context that students are likely to engage with. They are some of the key things that we want to engage with. And to approach this, some of the ways that I like to consider it, which you may want to speak to your PPG students about, and they in turn may want to speak to students in schools about, I find it helpful to consider my relationship with AI the same way that I would consider my relationship with other people in a professional role. And I'm going to do a little bit of stereotyping and generalising here just for simplicity. But am I treating AI the same way that I would a supervisor where I treat AI almost as the expert and just run with whatever it suggests? And this one's greyed out of it because that's very much not how I would personally use AI. But you may have found is sometimes the way that your PPG students or particularly kids in classrooms might default to using AI as though it is the expert. I tend to use AI particularly for numeracy purposes in these other contexts as either a colleague who I might use for different perspectives or ideas of feedback, but who I then decide which of those I actually particularly like and want to continue with myself and which ones I don't actually want to take on board. but particularly again, might have a broadness of ideas that I haven't quite considered yet myself. So just broadens that thinking a little bit. The one I particularly use it for, that we'll see quite a lot of, is more of an assistant. But again, in the sense that... when I have AI as an assistant and I assign it a task that I think is within its capability, I'm still in control over reviewing that work, deciding if it's fit for purpose. The other role I might add on to that, that I sometimes use it for, is probably akin to a contract worker. which is tasks that I have limited skill in myself. And I'm quite cautious of this, particularly in a numeracy context, of I don't really want to contract out the numeracy element, because that's the element that I should be the expert. And HUM. And for PPG students, for kids in schools, that is where we actually want to work towards upskilling ourselves so that we have strong numeracy capability and are able to critically think about some of these other things that come back. But one example of where I may have a bit more limited skill is things like graphic design. You know, if I want effective posters that I can use to talk through different strategies and frameworks and things like that. I know my personal graphic design ability. is not as strong, whereas some AI systems can do that very, very well and very, very quickly. So that's the sort of task that I might actually contract out. And then again, I have that other side of the thinking about, again, evaluating in both of these assistant and contract worker contacts, what's fit for purpose, what I might want reattempted with a bit of clarification, so that refining of prompts, for example. Also, what might I want to entirely reassign of if it looks like, you know, similar to getting an assistant, this is actually beyond the capabilities of the AI to do. What do I instead want to say? Well, don't do that. I'll look at that element. of what I'm trying to achieve and instead have something else to look at instead. And so with that problem we have with the two different bags of rice, that might be an example of something that I don't want to keep trying to use AI for because it's clearly not. actually going to approach it in a way that I think is fit for purpose and I might look at other ways that

I can use AI effectively in that context. When it comes to, again, your perspective and feedback and ideas and things like that, again, I still critically think about which ones I do, I don't want to take on board or pursue further, and which ones I disagree with, or I'm just not interested in pursuing. And again, we'll see lots of examples of this. as we start to work through the concepts. Um, and so often what we will see is this backwards and forwards relationship of what I do, um, of output in a prompt. I'll get AI to generate something. I'll look at what's come back. I can evaluate it and decide if it is what I want to actually go forward with. I'll refine it based on that professional expertise that I have. until eventually again, I'm happy with what I'm going to use. And again, that concept of ownership that I believe Craig's talked about, that whether we do or don't use AI as part of our process, ultimately it is us. That owns what we end up using. Now in reality, that refining and evaluating process can often happen many, many times as we work towards something that we're eventually happy with. Sometimes as part of that, as you've already seen a few examples of, Rather than staying in the same chat with an AI agent, I might actually start an entirely new chat again with a better prompt, and some of you may have experienced working with AI agents that often, if you keep going, trying to refine in the same chat. You start to reach a point where the results don't get better and sometimes even start getting worse than what you started with, and that might be the time to switch to something new with a better prompt based off of what you've learned from its responses. There are some times, again, where you may find that once you've evaluated what comes out, you actually make the choice that this is beyond an AI's capabilities and you don't want to use AI for that purpose either. And there are some cases where after going through this whole process, you will have come up with something so good that's reusable that you may actually want to engage in some of that meta prompting of asking AI for what we often call a handover prompt. Okay, so If I found a process particularly fruitful with AI, I might ask it to suggest to me what prompt I could have used way back at the start of our chat. to get the results that I wanted so that I can actually use that as a starting point in future to much more quickly get out what I want. And again, we might explore some of those different examples as we go through. But again, to Amelia's point, there is a balancing as we do this about where you're investing your time and effort. And particularly balancing that also with how highly we rate AI's likelihood of producing something that's going to be effective in that context, and also how important it is that AI gets it right. And so, there'll certainly be some instances where, you know... near enough might be good enough for what I'm doing. Okay, when it comes to some of my resource generation, that I can get a resource that is good enough to go in front of a class, you know, maybe could still be refined, but doesn't need to be refined further. in order to be effective. There are some cases where I want to keep pushing to get things pretty much perfect. And so I start to consider again those elements of context, different intentions that I have, and start to engage in some things that we've already looked at of, am I going to try and give it a role? Am I going to do some meta prompts? And as we go through the session, we'll also look at some other useful tools like asking if it wants anything clarified or asking it for multiple different options that we can choose from. HUM. Okay, we're doing well. I know it's a lot of content that we're working through. We are now at the good turning point of the session where we're going to start getting really practical. So from concept 2 onwards, concept 2 and concept 3, the rest of this whole module is now going to work towards every single person in this session actually developing their own numeracy task, including supporting resources. but drawing on some of the things that we've established so far in this session. Uh, with a prompt. Lovely. I'm just noticing some things in the chat that I just want to cheque first. Wonderful. And of course, I've noticed one person's coming to chat. If at any point you

do need to step out of the session for prayers or for any other reason, of course, please do go ahead and then rejoin when you can. We will take a break quite shortly as well. But please, obviously. Very important that if you have dedicated prayers or if you just need to head off to the bathroom or something like that, please do feel free to do so. HUM. Okay. So this is where, again, we are going to draw a little bit on potentially some of your... pre-work from the online session. Let me just work back through to... Where that was. HUM, so. Hopefully, in the online session, one thing you did do. is start to think through the elements of a prompt that you might use, specifying a little bit more about your class context or a typical context that your PPG students would be working in themselves. So if you imagine you have one of your specific PPG students, and a typical class I might go into. Um, and those elements that you would consider for an effective neuracy problem, and the sorts of approaches you might. Take building up to again, hopefully many of you have already done a strong example of a numeracy task prompt to put into AI. What I'm going to do is I think this is probably a particularly effective point to in a moment we will take a break and we'll do a couple of things. So In a moment, we'll have a bit of time to certainly have a break, stretch our legs, get some more food or drink or whatever we need, and have a look back at if you got the chance to do this online component, what you got out of it. and what your evaluation of the output was. If you didn't, it's going to be a little bit of opportunity to try and do a quick sort of rust job of putting together a prompt yourself. And then being ready to have some conversations around what was good and what you got back and where you think could be improved. HUM. And the way we'll do this also is I'm going to start to use some breakout rooms as well. So in a little bit, what I will try to do, oh, actually, can I still? There is. I'll put us a little bit later into some rooms of two or three of us, just so that we actually have a bit more opportunity to each discuss with some people what our findings were in this, and then we'll rejoin together. Otherwise... with 40 of us here together, we don't all get a chance to really have a voice in here. So what I'm going to structure this time-wise is... Let's have probably about at least a five-minute entire break, and then I'm going to put us into some rooms for about 5 minutes just to talk with each other about how we made our different prompts and what we thought of the outputs of them, and then we'll come back together as a group. We are going to have another break in a little bit as well, so then it's not going to be just this short little break, but hopefully at least some opportunity as well to step away again, get a drink or go to the toilet, or... start to think about how you might design your prompt. I'll stick around here as well if you've got any other questions in the meantime. And I'm just going to turn my camera mic for a moment, just so I can also do a bit of a stretch, but I am right in the room. So if you put something in the chat or speak, I will hear and I'll see you. Okay, so I'm going to try and put us into some breakout groups now. The people that you're not paired with may still just be on break for another moment or so. So just be mindful if someone doesn't respond quite yet. that that might be the reason why that's the case. But this will just again be a bit of an opportunity to cheque in with someone else in the cohort about how you've gone about crafting your prompts, what you are observing in the output that comes from AI, and whether you think... what did come out was an appropriate task, whether you think it needed some modification, and if so, what those might be, or if you don't really think it's appropriate at all. And when we come back together, we can share a couple examples of those and talk to them a little bit more, and then see how we might improve this process in order to be able to generate better numeracy tasks. And I am specifically going to look at one in the literacy or English context as well, because I know again that was requested earlier. So I'll have a go at trying to kind of the prompts to look at that in the meantime as well. So let's go into our breakout rooms for probably not too long,

somewhere between 5 or 10 minutes, just having a bit of a discussion with each other before we come back in a little while. HUM.

[1:59:37] Participant: Do I have a freak out friend?

[1:59:42] Blake Segler: And let's see. Should take just a moment. Thanks to work, just generating them now. HUM. So hopefully that should be on your screen, the option to join a room that will have one or two other people to have a bit of a discussion with just for the next 5 or so minutes.

[2:01:31] Participant: Sorry, Mister Blake, is there another room that we have to split, or just this room?

[2:01:37] Blake Segler: Uh, it's I sort of come up in your scheme. Let me just try again. HUM. Okay, hopefully that. Should work this time. It might just take one moment or sorry. HUM. So. Okay, I think there might be an issue with, depending on the version of Teams that you've got, I think it's only letting certain accounts actually go into breakout rooms. So you might need to just rethink that. I'll have a thought myself about that. Bad. Let me see if I can try and fix this. Otherwise, we can more generally do some discussions in here as well. I was going to say, bye. If I can fix that. Mhm. No, I think yeah, certain account types, it's just not letting you use the breakout room functionality. That's not a problem, though. We'll modify slightly some of the places that we're going to use that for the rest of this session is perfectly fine, and I'll I might just bring... the rest of the cohort back at this point. And then we can have a... conversation all together. I'm just going to close off the ones that did assign and hopefully it might all be back together. Anyway. Okay, I'm just gonna give... A brief moment I saw, wow, people. re-enter the main group. And apologies, it does look as though the breakout rooms aren't working for everyone based on the type of Teams account that you have. So I'll consider if we do still try and use them partially for the rest of the session. Or if we sort of abandon that and just use a different way of working together for the rest of this one. You know, trying to give a little bit more opportunity for us to actually each have some conversations around some of these elements. HUM. If anyone did want to give any comment about for their own individual prompt, what they thought of the output. Please do feel free, again, even in the chat or raise your hand. Otherwise, in a moment, I'll start to model a little bit of this with the example one that I did. And we can start to look at how to improve what we're putting in and get some better options coming. Out. But I give a quick mind if anyone did want to comment or feedback or ask questions about some of things I did observe and what they might like a little bit of input on. Now, my one, as again, you would have seen in the online example. So I gave a bit of context to mine and again, this is a little bit of the balancing act. I've told it. An example class that my pre-service teachers might go into, like a year 7 science class that's doing biological sciences at the moment. I've told it a bit more about the context of a North Metropolitan region and a bit of a student makeup. And then what I'm hoping to get out of it. So in my case, looking at newly problems that are relevant and meaningful while aligning with the curriculum content, I want that opportunity for differentiation that all of my students hopefully can address the task. but with different levels of sophistication in their response. And for me, again, thinking about some of the other ones we've discussed before, making sure that the problems open enough that they can make some of their own choices regarding their approach and assumptions. Okay, so making it again, much more real world, much more engaging in critical thinking, not just A Mats problem. And then we start to look at again some of our approaches and the potential assessment that we're going to do. And again, in my context. There is a framework that's often used in Australia for problem solving. So I've chosen that one. But

you could align it to something like, again, your poll use problem solving steps for those familiar with that framework, which you may have again seen if you've done the online component already, you might simply align it to your different elements of numeracy that you see in your national assessment. So the knowledge, the context and the cognitive processes that go into approaches. approaching numeracy problems that we'd particularly be looking for, both in student responses and in what we'd want to develop. Uh, and so, and I got back our response, um, and it wasn't. bad, but it wasn't particularly great either. I found it a little bit too demanding of students. I found a lot of the maths unrealistic. There was probably a little bit too much reliance on maths and not enough on the science. It also didn't use the correct curriculum for me. So Australia's a bit interesting in that while we have a sort of national suggested curriculum, every state and territory actually then uses that and comes up with their own curriculum. So they might use it as is, but they might change it, or they might do something completely their own. and since use the wrong curriculum for my actual state as well, and a few other limitations that I found. And so that's where I then go and look at the prompts that I had originally again and think about ways that I might change that. And there's a few elements of this that we might discuss. I'm going to, I've pre-generated some of these just so that we don't suffer a little bit of the time that it takes AI to think through them. I will talk through just some of the small modifications that I've made, and then I really do want to try and also look at that English prompt, and we'll see what comes back there, and then we'll push forward a little bit with what we're gonna do with those. So, one of the easy... Once. is, again, I noticed that it actually used the wrong curriculum and that's quite an easy fix for me. I actually downloaded all of the curriculum documents for my state, Western Australia. I did both the mathematics curriculum and the science curriculum. Because they're going to need to consider both. What mathematical content knowledge do students up to, again, in my case, year 7 is the example I gave. And what science content are they focusing on in that year if I'm going to align it. So I've given it that. Extra context. HUM and told. and that I provided those documents. I've gone on to specify a little bit more as well that when it looks at valid approaches to the problem, I want it to be sure that it's only really drawing on content knowledge covered in the years up to the class that I'm looking at. Because again, I did find from the output that I saw, some of it was too advanced and inappropriate for learners at that level. And more so, I specified that I don't want it to look at any general or specialised knowledge beyond what I would expect of a typical Thirteen-year-old, because I did find again the example that it came up with. relied on too much knowledge, understanding, cognitive ability that I think year 7 students, which would be about 13 years old for us, it would be beyond them. I've kept the rest of that, and then I've added a few additional things. One is, I've given a little bit more constraint around how long I'm looking for. I've suggested, because I'm quite keen on this activity, I did two 60-minute lessons. You might want to do less. So up to you what duration of task you're after, but it helps to decide going in. how long you want the task to pay out for in a typical class, because that's obviously going to dictate the scope of what comes back. Again, looking at some of what came out and the considerations I'd probably have, which will also speak to that consideration of what, you know, why is it relevant that we're covering these skills? I did want to specify as well. that in this case, where I was looking at a biological sciences class, I didn't want the task to turn too much into a diversion into mathematics alone. I wanted to make sure that the scientific concepts that as a science teacher I'd be focusing on, they were playing a central role. and when not just, you know, what we sometimes call window dressing or a superficial add-on to what would really just be a mathematics problem. And one of the really big things that helps as an advice strategy is rather than just asking for a

nearest problem, I'm actually gonna ask it to give me ten of them, because what I'm going to do is I'm going to quite quickly look through the list and I can be much more definitive of just throwing away. suggestions that it comes up with that I don't particularly like because I don't think they're appropriate, whether that's because of the level that they're at or the relevance of things like that. And much more whittle this down towards a short list of ones that I think I could quite quickly fix up. because they're close enough to being good, or potentially some that I think are good straight away that I can use with very little modification. And that's where, again, I think AI is quite useful in terms of, I mentioned before, coming up with ideas of what could be relevant is I might have to think for a while of lots of different meaningful contexts. To put this in. The AI will do this almost instantly. This one when I did it, you can see it reasoned for one single second before it comes up with its list of 10. In reality, what I'd probably also do is I wouldn't just put it into the one AI. So this one is CodePipe that I've got up. I probably also put it in Claude at the same time. I put it in ChatGPT at the same time. And of all the different ideas that comes back. I don't need 30 good numeracy tasks. I only need one for this topic, maybe a couple, depending on how long this topic goes for and how many opportunities I want to look at numeracy throughout it as a science teacher. And so I can very quickly get all of these different ideas. and then just reduce it down to my real front runners of what's going to be the best ones to go forward with. What are the most relevant? What are the most appropriate? And so I'd look through. the list of 10 and decide, you know, for a wildlife friendly school garden, do I think students care about this? Do I think the numeracy is at the right level? Do I like, as a science teacher, how well it links in with the content? and a brief assessment on that. Do I, what do I think of looking at how many insects live in the school grounds? And across the 10 different examples that I've got, Which one do I actually consider to be the best one that I'd like to move forward with? That's the general idea. OK, and so. What we're going to do now is try to refine our prompts from before. and see if we can generate a list to sort of work through. While that's going on, I've started the English one. HUM. Now, you'll again... make this appropriate for an Indonesian context in particular. I've mostly copied what I used before, but I've started modifying it for English. So now I've got to consider a high school English class, and I've got them currently studying persuasive text. And those of you that do have more of an English or a literacy background may want to do this a bit differently than I'm doing it. I've kept the other factors the same for myself, but you might want to modify them a bit more for your context, and so I want to provide students with a task. that incorporates numeracy, relying on their personal numeracy to not only evaluate claims, but to consider how arguments can be constructed in order to persuade the reader towards a particular. Keypoint. Really all students are able to tamp on by different levels of sophistication. And. And again, I might give curriculum documents, I might not. I'm going to remove this sentence just for the purpose of this. And I'm going to probably think a little bit more in this case. I might need to consider what is more relevant towards an English context as well. HUM. Sorry, students. Might consider how? Uh, genre context. Audience. and purpose contribute to the way in which HUM, numerical arguments. Have been used, HUM? and whether or not the... Argument produced by the author. He is convincing, or where they may be. Sceptical. And. Multiple valid approach shows. To the task. Should draw only on mathematics context knowledge that students will have covered. I can keep that the same, probably. HUM. I'm going to remove this bit for the mic because that's a bit WAE specific. Let's ask for 10 appropriate contacts or tasks. That. Students could explore. It should be achieved by the cost of. I understand 30 minutes. Lesson. OK, so that's... Not too much involved in that prompt. Let's try and see how it goes in here. Okay, and so again, very quickly, I've got 10

different ideas. of what the stimulus might be. It looks like, again, bandwidth-wise, it's had a little bit of trouble generating some bits of it, so ignore some of the weird type of type things that have come up. But now I can decide as an English or literacy teacher, which stimulus do I think might be particularly relevant? What, again, elements of this task I particularly do or don't like? Do I want to look at energy drinks or school canteen prices is always a good one. Again, do I... like some of these opportunities that are discussed to the right. And which ones do I think students are likely to be engaged in? I know for the Australian context, some of these students would really care about. and some of them. I don't think students would care about it all and not get rid of. I, you know, I think students in my context, sadly, will not care as much about a recycling poster. as they might about canteen prices, which students in Australia constantly talk about. students probably care about whether or not phones should be banned at lunchtime, but the reality is we actually have a state government mandate that students aren't allowed to use phones. So it's possible that I could do this in terms of a persuasive argument to change the state government's mind about that, but I've got to make sure again that that's contextually meaningful to me, that I'm not suggesting that aside from a change in the government's position, that's not something that's going to change within our school. And then I think about, again, some of the opportunities that I've got. If I want to look at something like misleading graphs, that in Australian context, we see a lot of misleading graphs in the media, okay, that change the axes and things like that, or do pie graphs. that don't actually have the right proportions of each segment. So can I think about some of these elements that I might actually want? And again, mostly I'm just actually looking through this list. and picking what are some of my favourite ones that I think I might assume. That's the process. What I'm going to get us to do over the next. let's say 10 minutes or so. Certainly if you've got questions, ask questions, but I do want to make sure that if we haven't already done so, that we have a bit of a go at doing this ourselves. So again, think about a specific context that you would want your PPG students So, I can set up. Write a bit of a prompt that looks at describing what that class context might look like, what elements of numeracy you'd be particularly interested in students demonstrating, and asks for a number of tasks. You might want to just do say 5, because for the time that we've got together, it might be a little bit much trying to review a set of 10. but get it to produce say 5 different tasks and then think through and decide which one of those you actually like the most and why and what modifications you still might want to produce. I will copy into The chat as well, for what it's worth. what my modified prompt looks like back in the science context. I'll also put in what that English prompt looks like. You're welcome to use either of these as some starting points or use what you did back in the online section as a starting point. are entirely stopped from scratch. are entirely up to you. But yes, otherwise take the next 10 minutes or so to try and do a bit of a revised prompt. Get out. some different possibilities. and then actually decide which one you like the most. And if you have questions or you want a bit of help troubleshooting somewhere over that 10 minutes, please do also ask me some questions during that time as well. But otherwise, we'll use this as a little bit of Independent working time. You can ignore. HUM. this little bit about breakout rooms. We won't use that one for this task. But in place, they clarify the way if you need. Stupid. Okay. HUM. Now we'll come back to that in just about a minute. I can see some people have jumped into starting to write into a pallet where you can share the posts, sorry, the tasks that you like the most. So do feel free if you'd like to do that. Don't worry too much if you don't get a chance to add to that. But it will just give a little bit more opportunity again to just share some different ideas and justifications for the choices that you're making. but perfectly okay if you just want to stay focused in this session as we

move forward. Uh, and we'll again just talk through some of the general considerations that we'd apply. HUM. That's the timer. HUM. And so hopefully what you'd like to notice is that there is a big difference in quality between your best suggested task that you would have gotten with the slightly refined prompt and particularly when you do have that openness of asking for five or 10 different potential tasks compared to when you just ask for one task and what you get. So it gets us a much better starting point of what we want. and a much better way forward. What we're going to now look at for the last part of this session. is then really fleshing out that task and equipping it with a lot of the different resources that we may want in order to teach this effectively. And particularly looking back, some of those ones that may have otherwise taken us a large amount of time to develop ourselves. And so this is where, again, we're thinking through and evaluating the different things that have come back. So if I model a little bit with that English example, because I do like that suggested context, but because it will take me a little time to generate more resources with this, I might jump over to some pre-prepared ones. Otherwise. So I asked it just because I like the context, mostly to look at the canteen price rise. It might not have actually been the best one in there. So some of you may have chosen something different. But then it generated this for me of looking at... two different ways of presenting this price rise, one that the canteen wrote, one that a student wrote, and to discuss which I think is more convincing and why. It's not a bad task, I think, for an English purpose, at least to start off with. Again, we can talk about the reasons why the way the canteen presents this might be different to how a student presents it. We can talk about, again, some of the choices I've made in general with regards to things like language, but also those numeracy elements of You know, the canteen's choice to say only 50 cents versus the students' choice to say 12.5% more. But I might look at that and say, there are things that I still want to improve in that. So, for example, when I'm looking at it, I might decide that for a high school student in an English subject class. It's a little bit short. There's not much textual analysis I'm really doing there. So one thing I might ask it to do is to change these to each be say 250 words. I might look from a numeracy perspective alongside and say, So only one numerical element to this, could I get some more things in there? And when I look back to some of the other suggestions that came up with... Some of these talked about, again, numbers of students surveyed and how that was represented. So I might ask it to put some of those elements into my texts as well. So there are multiple things that students could draw off of. I may ask it to ensure that there's some competing elements of argument towards this. And then I may make some more personal choices around things like, do I want to tell students who wrote which of these different pieces of text, or do I want them to try and suggest who wrote them and why that might be? Do I want to actually again give them just the raw information, for example, and suggest ways that they might write it themselves if they're going to produce a convincing argument in a certain direction? Lots of different elements of how I might actually play around with this task. before we go too far and refine it a little bit more. I'm going to, for the resource development, I'm going to jump over to the one that I had generated beforehand. Now, I wasn't too big a fan of. the original numeracy task I have for my year 7 biological sciences class. In this one... When I actually get to it, I think this is the start of the prompt. Generated a lot for me, I must say. I don't know this is still the comparison of them. So I got it too, as you may have seen, in... The Paddler. I got it too. Oof, it's a lot of response that it's given me. develop what it called a frog shelter task into complete two lesson activity. And this is partly where again, because I put a lot into my prompt, it's given a lot out. So I asked for a clear framework that it could work in to guide students numerously thinking. I've asked for a lot of elements that I want it to include. And so it's

given me a full curriculum rationale. It's presented a scenario. This one, again, from a science perspective and the numeracy perspective, I like a lot more. So it's given me a lot of different factors. that we've got some data on and competing factors that students could consider and decide which ones they think are the most important. So the idea is we've got, what 125 has eight different sites where we might produce a frog shelter. We've got some monitoring data on all of them. And now students actually have much more open choice about which factors they're going to consider, how they're going to compare them, how they're going to arrive at a conclusion. And it goes through and structures different things that they might do at each point of this, and again. I'm going to consider within that elements I might change. So the first thing I did is, well, I like the idea. I wanted it to reconsider some of that data, because at the time that I was written, when I looked at it, there wasn't anything that wasn't important, and I want students to be able to actually not assume they have to use everything. Particularly, as well, I wanted to ensure that multiple students are likely to arrive at different conclusions, but with good reasoning, so that this was a meaningful task and not just one where they assumed there was some one definitive answer. that I have as the teacher that they're meant to get to, that whatever reason they come up with would be potentially appropriate. And so, it did that and it came up with new examples, and once I was happy with that, don't worry too much about all of the details of this. That's when I started getting some resources. OK, so one thing I asked it for... was a task brief for students. So a clear single A4 page for students. That was a one-page problem brief, tasking with finding the best site. need to have the diet and explanatory notes, ideally a bit of a map as a visual for students to look at. I wanted to make it clear that they didn't need to use all of that data and that they should consider what they're going to use and how they'll ultimately make their decision. I also, because we're now getting into more important things and I'm getting closer to what I might want to have as usable product, I specifically asked, which is often a good step with AI, if it wants to clarify anything with me before it begins. And that's a very useful sentence to add at the end of any prompt that is potentially important and potentially time consuming. is let it identify if there's any decisions that it feels you should actually give some clarification on 1st. And so then in what took again, very little time, 38 seconds though still in this case, but it would take me a lot longer to do. I get this nice. Clear page that I could present to our students if I want. This is where, again, I probably would evaluate it and decide what other things I might still want to change. Okay, wherever I think this map is a little bit rough and I either want it to be a bit more accurate or get rid of it entirely. I'm happy that this was called science and numeracy as a topic. But I do think that maybe this is too many things now on the page for students to have to worry about at a year 7 level. So I'm still going to evaluate that and potentially refine it. But just to sort of show off a few different capabilities, I've failed that. and thought I'd just get AI to generate a few different types of resources that we can see some of its capabilities before we think about how we'd evaluate and refine them. So, this is where I really jumped in. and said, can you generate a workbook for the entire task with relatively little scaffolding? But separately, I want additional scaffolds I could use for each key element that I could give to weaker students. I want additional extending prompts. that I might give to more capable students. And so again, lots of people commented on this element of differentiation and even just resource development. of in, this one's taken a minute and a half, you can see. It'll take a minute and a half to get a largely usable workbook for students. That would otherwise take me a while to develop, as well as a whole bunch of differentiation materials that I can use as well. And so, in a minute and a half. I get a nice structured place where students can work. It's used the framework that I suggested back at the start that I like

my students to use because it aligns with Australian frameworks. Again, you might choose to do something different though. It includes a little submission cheque for students. But then it's also got. a much more scaffolded version that I might provide to students as well. So many more sentence starters, many more ways it's going to guide students about the different considerations they should be thinking about, rather than the much more open task where they've got to think about all those considerations themselves. Not all of them, exactly the way I necessarily present them. And one thing I might want is to be able to have these as separate documents, for example, so I can choose which ones I give to students when. But again, very quickly, an excellent starting point that would have taken me so much time to develop myself. And now I probably just need to either give a couple of refining prompts too, or go through and start editing. I've also got all of these extending prompts. Lots of different directions that I can push my more able students to start to explore. That's still going to draw on their numeracy capability. Again, one minute and a half's time out of my day during which I probably just went and made a cup of coffee, I did a different task anyway. So in terms of my time and efficiency and workload. Absolutely brilliant. A few other things that I've got it to generate. In this one, I've asked for a self-assessment rubric for students, and I've told her I want the language to be student-friendly and include some specific exam elements of student responses that would be reflective achievement at that level. And again, this one took 50 seconds. I will say that this one, when it generates. Maybe a little bit simplistic. I can. So this is 1 where I might want to refine it a fair amount more, but it's still gone through my framework. Okay, it's given what I would say is relatively student-friendly language. What that might look like at the different levels, and again, an example of something that they might say that they could self-assess or peer-assess them against, and this also talks a little bit to that element of feedback that people are talking about. is now I'm actually allowing students to engage in self and peer assessment so that more students can readily get feedback. Again, I'd probably edit this a fair bit. I think it's trying to specifically only do one element for each part of my framework. Whereas in reality, there's probably quite a few things that go into things like selecting their assumptions or applying the mathematics and so on. But again, an example of how it can quickly generate some resources. And then I think the last or second to last month that I got it today was then actually generate. A complete example student response, OK, that we could assess. Together, OK, and so this is where, again, I'm thinking. If I'm going to actually get students in charge of their self or peer assessment, I might need to be able to model this and work through it with the class of can we have an example response that we could talk through together and decide how strong it's evidencing decent numeracy capability. What are the elements that are good? What are the elements that can be approved? 46 seconds for a complete two-page or so example student response. I've then asked it further, can I actually have an annotated version that shows how it will be assessed against the rubric? And so then, you know, if I was just going to give it as a supplementary resource, I'll use that as a little bit of guidance notes. It's going through and adding some comments about what's done well, what could be improved, and so on. HUM and. Almost all of these, I will say, when I go in and look at them. Then, not. Perfect. There are elements that I think could be improved and refined. And so this is where I'm going to start to do a few things. I'm going to use my professional judgement to again decide what are the changes I might want to make. And within those, I'm going to decide whether I want to tell the AI what changes I'd like it to make, or if I'm just going to go through and do them myself. Either is fine, and I might sometimes see how well AI responds to a further prompt to refine what it's done and decide if it's again going to actually be fruitful to keep going with refinement, or if I'm going to find it faster to actually just do that myself. I

am also going to consider a little bit in all honesty, as I mentioned before, when is near enough good enough of depending how I'm going to use this? OK, do I need the absolute? perfect, every single element of the rubric is as ideal as it could be, version. Or when I'm just using this as a teaching and learning opportunity to explore numeracy, is a rubric that is still relatively decent. going to be sufficient still to get students starting to at least reflect on ways that they can improve their responses. And there'll be a limit where I decide from a practical workload perspective. when I'm actually happy that something is classroom ready for the purpose I'm trying to let it serve. And so I can go through again and decide which of these elements I do want to build. I don't necessarily need all of them for every single task that I do. There are some other things that I might also want to get it to generate, like, say, some teacher guidance notes. HUM, and then consider again, you know. which bits are particularly going to support my students learning, which bits again I need to be quite protective over. What we'll notice throughout this is probably the balance of where the numeracy element and particularly the approaches and the reasoning is lying. We identified sort of at the start that we may land in trouble in terms of getting the AI to suggest a specific approach. What I'm often doing, again, is I am safe. you know, what makes a good numeracy task. I'm safeguarding the specific approaches students would take. But in terms of providing opportunities for students to engage in numeracy, and again, going back to that wonderful first question, About relevance for students to engage. I'm going to much more quickly align at a good task that aligns with my curriculum or meaningful real world context if I'm more generally exploring numeracy, with good opportunities for students to engage in lots of varied ways and actually apply that critical thinking and reasoning. and with a whole lot of additional support resources, whether that's templates or rubrics and things like that, and all of those are elements that again are largely. text-based administrative that AI can do generally quite well and that I can quickly have it refine or I can refine myself. In terms of then specifically looking at some of those specific numeracy elements and reasonings and how I guide conversation, Again, I can still hold on to that myself as a teacher. Okay, so I'm using AI a lot more to support me in providing strong opportunities to engage in numeracy. without trying to turn AI into the numeracy teacher itself and holding on to that pedagogical agency as the expert that I am. Now this is a good point probably to pause for any questions or clarifications on some of things we've seen so far. Before I again have a little bit of a go at some of these elements ourselves. HUM. And if you do have any questions, please do, ohh, accidentally started the timer, please do. Put them in the chat or feel free to raise your hand. Otherwise, what we're going to embark on in a moment. is again a little bit more independent working time to start to generate some of these, review and refine them ourselves. So in a moment, what I'm going to get you to do. Is spend some time with the task that you chose back a little bit ago, and I can see. some great examples from different contexts that people have added to the Padlet already. For that task, now start to get AI to generate some other resources that you think would help in facilitating that task. So whether that is things like templates, whether it's some additional differentiation resources of additional scaffolds or extending prompts, if you do want to potentially look at some self-assessment rubrics, or alternatively, if you want a teacher assessment rubric, which would likely have some different language again to a student version. If you want to look at some guidance notes and things like that, have a think about some of the resources you want. Probably try not to spread yourself too thin for this next bit. So start with one supporting resource. See what comes back. Have a look at it. Think of some of the key things that you do like in it. and some of the things that you might still want to refine. And again, see if you can give some of those additional refining prompts to guide AI into improving it. And again, you

might find that you reach a stage where, while you still want to refine it, It would be almost faster to just download the Word file and start fixing it yourself, rather than the back and forth with AI, and that might be where you sort of stop with that one. And so try and explore that for at least one resource. Maybe explore a couple of different things that you could generate with it and see what your initial impressions are. And again, I'm going to be available during this time. So if you want to ask some clarifying questions of things that aren't quite going right, please do. But we're going to spend, oh, time's flying by. So still spend about 10 or so minutes just trying to start embarking on this process of generating, evaluating and refining some supporting resources. If you want again, but... Don't worry too much. There's another place where you can start to record some of these to just sort of compare. within the group. Uh, so I'm looking at... a specific resource, some of the things that you have done to refine it and where you might keep going. But again, because time is a bit limited, I don't worry too much about recording this just for the moment. Spend a bit of time actually just exploring with AI and asking any questions. That do arise.

[3:07:50] Participant: Can I have a questions?

[3:07:52] Blake Segler: Yes, please do.

[3:07:53] Participant: Sure, go ahead. Yeah. My question is, as an English teacher or English instructor, then do we need specific training to teach numeracy concept or is there basic numeracy knowledge enough to integrate numeracy into the English lessons? Let me freeze that neatly.

[3:08:15] Blake Segler: That's an excellent question. HUM. I would say it's a bit of a balance, to be honest, of, you know, the purpose of numeracy, again, we're looking much more at that personal capability and ideally we are able to model to students the way that we would engage with numeracy ourselves and particularly, you know, we have that advantage of, you know, we've talked again about that distinction between numeracy and mathematics. One of the really powerful roles that people that aren't in the mathematics subject area have to play in the promotion of numeracy is that personal experience of how does someone who is not of a mathematics background still go about applying their personal numeracy in real life? and in that context. And so being able to model that is very important. Now, that does still rely on a reasonable level of personal numeracy, and so there's the chance that, you know, for some people, and particularly amongst your PPG students, depending what their own backgrounds are, it is sometimes an area where people may find they still need to upskill themselves a little bit if their own level of personal numeracy isn't particularly strong, that they may still need to develop some of those skills themselves. So it's a really good opportunity to discuss with your PPG students the importance of working with colleagues. Okay, and so likewise, myself, who I would consider myself to have a relatively strong level of personal literacy, but my background is obviously more on the numeracy. side. Then I might have some collegial conversations with someone like Tamara, you know, who'll speak to you on Wednesday, about how would she approach certain tasks and are there particular things that it may be worth emphasising with students as I go through them. I'm probably not expecting a significant barrier myself, really, for my ability to engage with those. But part of that professional collegiality is, again, those sorts of conversations that can help. support students. It is an area where I could ask AI a little bit as well about what it would recommend, but this is where I would be cautious because now we're in the realm where I might not be able to effectively evaluate what comes back. And again, am I treating it now still as a colleague, or have I shifted to where I'm almost treating it as a supervisor and where I might feel safer, either exploring some more highly reputable sources, or at least being able to

evaluate what comes back. from AI and being able to ask it what its sources are and investigate those. So use as more starting points for developing my literacy on numeracy personal capability. Or again, do I just want to say Tamara works down the hall from me? She's super friendly and she owes me one because I did help her with a numeracy problem the other week. So I'm just going to ask Tamara what she thinks about this instead. And those are some of the considerations. But again, are really important to say in terms of that numeracy and literacy perspective, that distinction between mathematics and literacy and, you know, English and literacy in our context of, we're not meant to be going to high levels of mathematical content with numeracy. when meant to be using critical thinking and reasoning. And yes, the level of mathematical content knowledge that I'd expect to be useful to everyday adults going through real life scenarios and things like that, but not again a particularly advanced. level. Okay, thinking through more of those structured thinkings. And so again, when we look at... you know, elements of what we consider useful in numeracy. For me, again, I've broken it down in some of my prompting around how we might approach a problem through this understand, plan, do, consider and communicate framework, or talked about, again, the sort of critical thinking that goes into it. For you, again, you might draw on those elements of numeracy and the Indonesian curriculum of the content, the cognitive processes, and the context, or any other way that you would consider, particularly within that Indonesian context. What Do you actually consider important in students approaching numeracy problems and having strong personal numeracy capability? And then it is again looking at within your authentic context, how does that appear? So again, when I'm looking at English, and again, English, not my main content area, so you very much will be the bigger expert on that. But to my mind, again, I think about what happens when I'm reading through a news article online that's got things like statistics or graphs in it. What happens when, as a student, I open up TikTok video. I don't know if TikTok's a big thing in Indonesia, but you know social media video as a student that again starts throwing out some numbers at me to try and convince me to do a certain thing. How swayed am I by that? kind of text type, because in Australia, within our English curriculum, we'd also look at media such as videos and things like that. What are the arguments presented towards me? Again, what is the purpose that they're trying to achieve? Who is the author of those? And again, can I be critical about that? Can I recognise why they might be presenting arguments a certain way? Can I fact cheque them? Can I consider if I actually do think that is a compelling argument? Or if I might need to... So, I can really consider the limitations of what's being presented to me and what other research and representations I might need to explore myself. And so, again, looking within your specific context, again, to me. As a non-English expert, persuasive text of different types is a big one where I think this is somewhere where people use numeracy in certainly particularly motivated and sometimes misleading ways that I would want students to be able to engage in numeracy within that. Context. But you'd, you know, be aware of a lot of other ways in which, again, you receive those features within that learning area.

[3:16:32] Participant: Okay, thank you so much.

[3:16:33] Blake Segler: No worries. Now, I'm just looking at the time. I always struggle with time. It always seems to go by very fast, even for a relatively long session, and you've all done so brilliantly at maintaining focus and engagement. throughout this. So going to start to wrap up. I'll still be available once we get to 6:00 if there are more questions that people would like to discuss for a little bit longer. But let's start to talk through just some endpoints of this. I will also put my e-mail address into the chat afterwards. So if you do have further questions that you want to talk through, more

than happy for you to reach out via e-mail and ask them as well. But some things that you're likely going to want to reflect on is thinking about

[3:17:35] Participant: Yeah.

[3:17:37] Blake Segler: Okay. Sorry, just got microphones. That's great. I think I'm excited.

[3:17:40] Participant: Sorry.

[3:17:46] Blake Segler: Thinking again about where you found AI effective and where it was perhaps challenging. And again, you know, one of, you don't need to worry too much about this pilot, I think. I'll put the link in just in case people do want to contribute to it, but it's going to be particularly, I think, an after the session one. If you do, HUM, my... Big one that I tend to look at again is that efficiency of AI. of there are not many places where I expect AI to have more expertise than myself. There's certainly some. Again, when I'm looking at producing things like graphics, often a lot of AI agents are much better than that at me. For your PPG students, again, this might be different. And again, if I did this evaluation for literacy, as we sort of just discussed, you know, then there might be some elements of literacy where maybe I do think AI might have the upper hand. on me. But one thing for certain is that AI is a lot more efficient than I am at a large number of tasks. And when I reflect back on again the tasks that I sort of went through with you that I generated, The number of. relatively reasonable, if not high quality resources, that it could generate incredibly quickly. allows for learning opportunities that I simply don't have the time to produce myself. And I can reflect back to when I was a teacher in schools and particularly in some of my early years of teaching. For the Australian context, We have a large number of classes that each teacher takes, with not a large number of resources centrally provided to teachers. So in my early years of teaching, I was up very late at night preparing resources, and often I would have to make the decision of, I could imagine what would be an excellent lesson. But I might not have time to actually produce what I would need in order to deliver it. And so I had to make choices of where I was going to do, perhaps not what I really wanted to deliver, but what was all I really had the time to produce. With something like AI, it's much quicker for me to generate resources and refine them to the point where I've got these really strong lessons catering for different diversities of learners very quickly. It's also, and again...

[3:20:57] Participant: Mr. Blake, can I have just two quick questions, but based on my curiosity?

[3:21:00] Blake Segler: Yeah. Yes, so.

[3:21:06] Participant: Is that okay? Yeah. I teach English and I really want to integrate numeracy into my lesson. Yeah. But I'm worried that my students will think that. It looks like I teach mathematics a lot in my text rather than teaching English. So do you have any tips how to integrate, how many percentage of numeracy or how to teach English without too much on teaching numeracy inside it? And then the second one about the assessment, for example, I give questions to my students.

[3:21:27] Blake Segler: HUM. Yeah.

[3:21:45] Participant: Yeah, it consists of numeracy in my text, yeah. Then my student answer correctly about the literacy and about the numeracy, about the calculation, but in but in wrong English. So how am I supposed to assess it?

[3:22:02] Blake Segler: Ah, yes. Wonderful. Let's answer both of those. So the first one was the concern that your students are going to feel like that you're suddenly a mathematics teacher because you're spending all this time concentrating on numeracy. And my advice to that would be, as a teacher of English or literacy or the equivalent name would be in the Indonesian context. You

should not be doing numeracy tasks every lesson. Not even close. That's not your job. And so I can't tell you. the exact right percentage of times that you should be doing numeracy in the context of your subject area. But I will say with full permission, Not often. You know, at most, a couple of times across the course of a topic, I would say is likely. Maybe not even that. And this is where, again, we think about numeracy as a shared responsibility. of if across all of our different subject areas. We're all doing at least every now and then a numeracy task, and particularly where it is relevant. then she didn't see it enough in different contexts to appreciate the importance of numeracy and how it will be applied. And I would think probably about the genuine applicability of your subject area of, again, how frequently when you're engaging with texts, Does numeracy come back up? Am I going to show [name] a numeracy context into a novel? Probably not. Not many of the novels that I've read feature a lot of numeracy. And maybe I might come up with an example. Some interesting ones. I don't know if The Curiousness of the Dog in the Night Time is a popular book in Indonesia, but that's one that's been very popular in Australia that has some numeracy embedded in it. There's other books that use numeracy, but I'm not going to go too out of my way on that. But again, you know, when I engage with the media, I'm engaging with statistics and graphical representations and things like that. And those are where I might more meaningfully bring it in as an example of why it is relevant. And I'm doing that within again. what's important to my subject area. Okay, so again, as I sort of suggested, for my English learning area, the elements of a persuasive text and the choices that are made and how someone of a particular background is going to try and produce a particular argument for a particular purpose. That, at least in the Australian context, that is English and a very strong part of our subject area. And then I'm applying within that the sort of numeracy elements. And so when I go to assess that To talk to your second question. I'm still really looking at how that fits within my English context of how students are able to talk about the choices that are made to produce a specific argument, how someone of a different background or with a different purpose. might present those differently. So the same that I would with the words talk about how someone might comment on a particular event in a way that makes that look really good or really bad. I can talk about how someone can use the same numerical data to form an argument that is really good or really bad, or choose the type of graph that they use, you know, and perhaps you know the scale on the axes and things like that to make it look really good. are really bad. Again, with the midterm elections recently, you will see different electoral maps and some of them make it look like, geez, all of America is Republican voting, and some of them make it look like all of America is Democrat voting. And again, what are the choices that are made and why that we want to be critical of? But yes, we keep it relevant to our context. We do it authentically. We don't see things in. And yes, I will say not every single lesson should be a numeracy. lesson. You know, literacy goes a little bit more the other way, but we're almost always using literacy. I've got to say in one form or another, but the extent to which I strongly engage in literacy in a maths context, certainly. It's not going to be to the same extent every lesson when I do it that way either. But all of us as a collective across our different subject areas, just at least seeing it sometime in meaningful ways. That's how we sort of get there collectively. HUM.

[3:27:48] Participant: Okay, thank you. Thank you, Blake.

[3:27:49] Blake Segler: No worries. I'm going to try and quickly wrap up. I know we are a little bit over time. And so just if you've got more questions, just give me about two minutes or so to wrap up

and then ask them afterwards. One thing as you go forward that I would highlight as general practise again is decide

[3:27:51] Participant: Yeah.

[3:28:08] Blake Segler: when AI is still being effective for you, when do you want to keep using AI prompts to refine what you've got, and when do you want to properly take control and start editing a document yourself? But the other thing is, particularly thinking about... what the teachers role is going to be in using those of as much as I can turn all of this into 500 different Word documents that I could just leave students with. I actually do still want to think about the flow of my lessons, what we do as a group. and discuss what students do in pairs or individually, what things I want to model versus what things I want students to properly have a go with and then compare at the end, what particular approaches I am encouraging students to use. So all of those key elements. both in terms of the numeracy and general pedagogy and lesson choices as a teacher, there are still things for me to consider. But that resourcing, again, I can just get incredibly more efficient. And the other thing, I'll very quickly say, thank you everyone for your extended focus. is even just that ideation at the start, coming up with ways I could relate it to my learning area. To again speak a little bit more to your question is really, that is where I is particularly helpful is that might be something I am really good at but takes a long time. Or it might be something that I'm actually not that good at of thinking about different ways that I could apply numeracy or literacy to my area. And AI might come up with some ideas that I hadn't thought of. And again, I might look at those and I might say 7 of the 10 are garbage. they're not really appropriate or they're not relevant. All students are going to look at me and go, hang on, you're just trying to teach me maths now and that's not the point of our English class. But I only need to get out, you know, for a particular topic again, maybe one or two that I really like and want to progress forwards with. So if we are struggling a little bit. more with finding those meaningful contacts. That's where this can help as well. And so idea as we sort of push forward with this a little bit more is again, spend a little bit more time further refining some of your ideas. Again, let me just type my HUM. e-mail into the chat. in case you do want to follow up with anything. Do you also have a look though, again, some of these pallets that we've started on? Do have a look through them to see again some of the different things that your peers are doing to give a little bit more inspiration for how you can apply this. But if you need a little bit of further help and guidance, do reach out. I suspect as these modules go on as well, you'll also pick up a few more helpful ideas to push a bit further along as well. So don't feel you need to be 100% certain on everything. I just, yeah. But hopefully some good starting points. And again, that key consideration of, I'm just going to close off with, yourself in the driver's seat to really decide, again, what do you like, what you want to improve further, what's going to work for your context. But We are already a bit over time. Thank you so much for people who have stuck around. If you do need to leave, certainly please do and thank you so much everyone for all of your engagement and focus and discussions and the wonderful questions many of you have asked. If you do serve some more questions, I'm happy to stick around for a little bit longer, but I do know it has been a long session that you've all done so well in. So if you are at the point where you need to sort of take a break and log off, any further conversations we have will still be recorded, so you can also just jump in. and see anything you missed out on. Again, do reach out if you need, but thank you so much again for all of your engagement across the session. It's been really lovely working with you, and I will see you again before the entire programme finishes off.

[3:32:59] Participant: Okay, thank you.

[3:33:01] Blake Segler: Thank you.

[3:33:02] Participant: Thank you so much.

[3:33:03] Participant: Blake, thank you very much.

[3:33:04] Participant: I'll see you again, Blake. Thank you, everyone. Thank you, Blake.

[3:33:04] Participant: Okay.

[3:33:07] Participant: Thank you, Blake. OK, everyone. Thank you, Blake, and everyone.

[3:33:07] Participant: Thank you, Blake.

[3:33:10] Blake Segler: Thank you.

[3:33:11] Participant: Thank you, thank you, Blake.

[3:33:14] Blake Segler: Hi, Paul. Have a lovely rest of your day.

[3:33:15] Participant: I, I... Nothing.

[3:33:17] Participant: Can we take a picture?

[3:33:19] Participant: Thank you.

[3:33:20] Participant: Sure.

[3:33:20] Blake Segler: Oh, certainly, yeah. If you want to take a picture, do you feel free? I think we've just, we've lost a couple of people, but let me stop sharing my screen and they're not going to see all the remaining people.

[3:33:23] Participant: Thank you, Brett.

[3:33:31] Participant: K.

[3:33:32] Blake Segler: And if you'd like to take a picture, please certainly do.

[3:33:32] Participant: So. Wow. Thank you, Blake.

[3:33:40] Blake Segler: No problem. Thank you so much.

[3:34:00] Participant: HUM.

[3:34:11] Blake Segler: I can, if you do have any other questions, do feel free to just put your microphone on and ask any of them. Otherwise, I feel happy, then I'll probably head off in a minute myself too.

[3:34:13] Participant: Yeah.

[3:34:29] Participant: Thank you.

[3:34:36] Blake Segler: It does seem like we're probably all okay, but again, do reach out via e-mail if you need. Thank you all very much again for today and have a lovely rest of your evening.

[3:34:49] Participant: Thank you, Susan.

[3:34:49] Participant: Thank you. Yay! Susan.

[3:34:55] Participant: How are you, Rekha?

[3:34:56] Participant: Thank you very much.

[3:35:00] Participant: HUM.

[3:35:04] Participant: Why not? HUM. OK, we. I get the HUM.

[3:35:24] Participant: Tangle.

[3:35:25] Participant: Yeah.

[3:35:27] Participant: The. I was a blasting.

[3:35:38] Participant: Okay, will.

[3:35:40] Participant: OK. Yes, I bet you.

[3:35:42] Participant: HUM.

[3:35:50] Participant: Ha ha. And the same service. Yeah, yeah. ED.